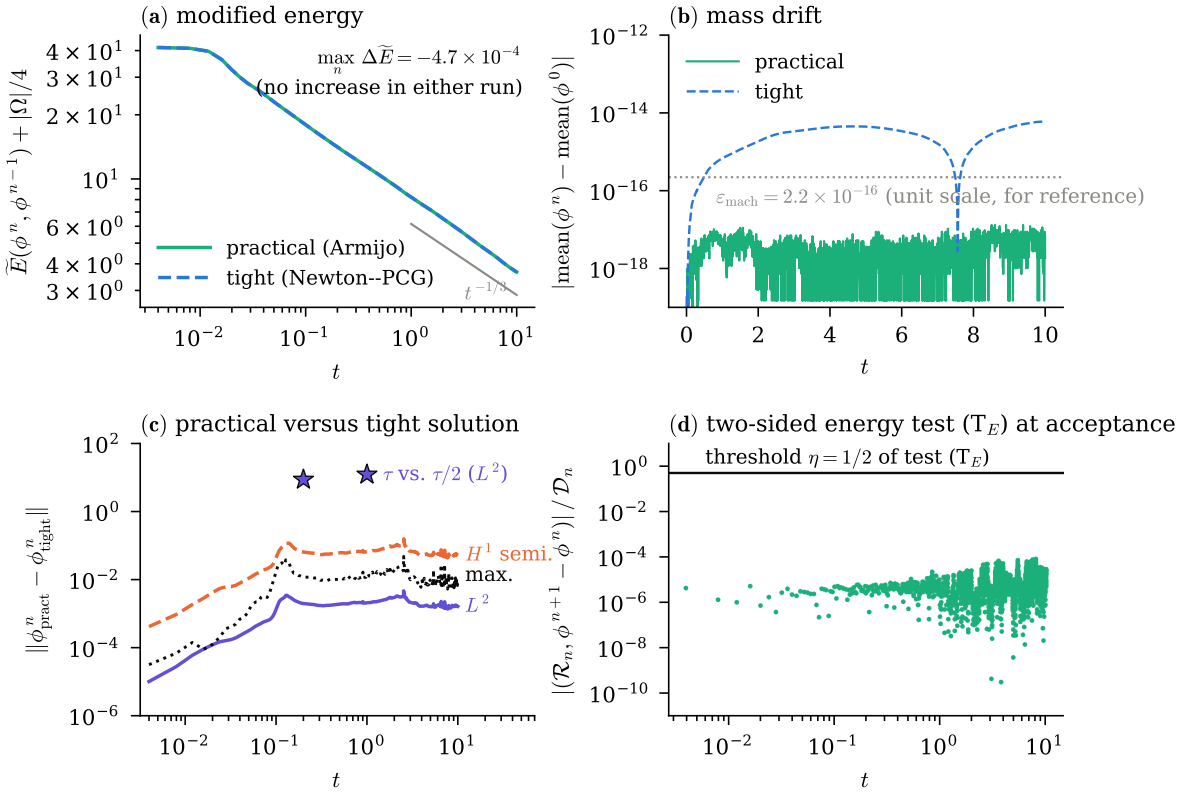

(a) modified energy
$\max_n \Delta\widetilde{E} = -4.7 \times 10^{-4}$
(no increase in either run)
practical (Armijo)
tight (Newton--PCG)
$t^{-1/3}$
$\widetilde{E}(\phi^n, \phi^{n-1}) + |\Omega|/4$
t
(b) mass drift
practical
tight
$\varepsilon_{\text{mach}} = 2.2 \times 10^{-16}$ (unit scale, for reference)
$|\text{mean}(\phi^n) - \text{mean}(\phi^0)|$
t
(c) practical versus tight solution
τ vs. $\tau/2$ (L^2)
H^1 semi.
max.
L^2
$\|\phi^n_{\text{pract}} - \phi^n_{\text{tight}}\|$
t
(d) two-sided energy test (T_E) at acceptance
threshold $\eta = 1/2$ of test (T_E)
$|(\mathcal{R}_n, \phi^{n+1} - \phi^n)| \,/\, \mathcal{D}_n$
t